# FEASIBILITY STUDY REPORT

## Face Mask Detection System

*Computer Vision Project*

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## 1. Introduction

MaskGuard AI is an intelligent face mask detection system developed using computer vision and deep learning techniques. The system is designed to automatically identify whether individuals in images or video streams are wearing face masks, providing a scalable and reliable solution for public health monitoring.

In the context of global health crises, such as the COVID-19 pandemic, the enforcement of mask mandates in crowded environments became a critical public safety concern. Manual monitoring of compliance is resource-intensive, inconsistent, and impractical at scale. MaskGuard AI addresses this challenge by leveraging state-of-the-art machine learning models to automate detection in real time, enabling deployment across hospitals, offices, airports, transport hubs, and other high-density environments.

## 2. Problem Statement

Traditional approaches to monitoring face mask compliance rely entirely on human supervision, which introduces several significant limitations. Manual monitoring is labour-intensive, susceptible to human error, and cannot operate continuously without fatigue-related degradation in performance. In large or high-traffic environments, deploying sufficient personnel for comprehensive coverage is neither practical nor cost-effective.

Furthermore, inconsistent enforcement can undermine public health goals. An automated AI-based system is therefore necessary to provide uniform, continuous, and accurate compliance monitoring without human limitations. MaskGuard AI directly addresses this need by integrating face detection and mask classification into a unified, deployable pipeline.

## 3. Objectives of the System

The primary objectives of the MaskGuard AI system are as follows:

- Detect human faces in both static images and real-time video streams with high accuracy.
- Classify detected faces as 'With Mask' or 'Without Mask' using a trained deep learning model.
- Provide real-time detection capability suitable for live CCTV and webcam feeds.
- Improve safety monitoring in public spaces, workplaces, and critical infrastructure.
- Deliver an accessible user interface enabling non-technical users to operate the system with minimal training.

## 4. Technical Feasibility

The MaskGuard AI system is fully implementable using currently available, mature, and well-supported technologies. Python serves as the primary programming language due to its extensive ecosystem for machine learning and computer vision. The system employs OpenCV for image preprocessing and real-time video processing, while deep learning model training is powered by TensorFlow and Keras using the MobileNetV2 architecture, a lightweight convolutional neural network optimised for performance on resource-constrained devices.

Face detection is handled using Haar Cascade Classifiers from OpenCV, a proven approach that balances speed and accuracy. Google Colab provides free GPU-accelerated compute resources, making model training accessible without dedicated hardware. The table below maps each system requirement to its corresponding tools, libraries, and resources:

| System Requirement | Tools / Resources / Libraries |
| --- | --- |
| Image preprocessing & enhancement | OpenCV, NumPy |
| Face detection | Haar Cascade Classifier (OpenCV) |
| Feature extraction | MobileNetV2 convolutional layers (Keras) |
| Deep learning model training | TensorFlow / Keras (MobileNetV2 backbone) |
| Dataset handling & manipulation | Pandas, NumPy, ImageDataGenerator |
| Real-time interface & demo | Gradio / OpenCV VideoCapture |
| GPU acceleration | Google Colab (NVIDIA T4) / CUDA |
| Performance evaluation | scikit-learn (classification report, confusion matrix) |
| Visualisation & reporting | Matplotlib, Seaborn |

The combination of these tools provides a complete, production-ready stack. OpenCV's efficient image handling pairs with MobileNetV2's pre-trained ImageNet weights to enable transfer learning, significantly reducing training time and data requirements. Gradio offers a browser-accessible interface with minimal development overhead, ensuring the system is operable by non-specialist users.

## 5. Operational Feasibility

MaskGuard AI is operationally viable for deployment across a wide range of real-world settings. In healthcare environments such as hospitals and clinics, the system can continuously monitor staff and visitor compliance without diverting nursing or security personnel. At transport hubs including airports and train stations, the system can be integrated with existing CCTV infrastructure to flag non-compliant individuals in high-throughput areas.

End users interact with the system through an intuitive web-based interface, requiring no specialised technical knowledge. Administrators can upload images or activate live video feeds, and the system returns annotated outputs with bounding boxes and classification labels in real time. The lightweight MobileNetV2 architecture ensures the system can run effectively on standard hardware, making operational adoption both practical and affordable.

## 6. Economic Feasibility

MaskGuard AI presents a highly cost-effective solution. All core libraries and frameworks—including TensorFlow, OpenCV, Keras, NumPy, Pandas, and scikit-learn—are open-source and freely available under permissive licences. Training and experimentation are conducted on Google Colab, which provides free GPU access, eliminating the need for dedicated hardware during development.

Deployment on existing server infrastructure or cloud platforms (e.g., AWS, Azure, or Google Cloud) incurs only standard compute costs. Compared to the recurring expense of employing dedicated compliance personnel, the automated system delivers a compelling return on investment, particularly in large-scale deployments. The project therefore demonstrates strong economic feasibility with minimal upfront and operational expenditure.

## 7. Schedule Feasibility

The project is structured to be completed within a standard academic semester. Development is organised into five sequential phases:

(1) dataset preparation and augmentation, (2) model architecture design and implementation, (3) model training, validation, and hyperparameter tuning, (4) interface development and integration, and (5) final testing, evaluation, and documentation. Each phase is achievable within two-to-three weeks, making the complete project deliverable within a twelve-to-sixteen week academic timeframe.

The use of transfer learning with MobileNetV2 substantially reduces the training phase duration, as the model benefits from pre-learned visual features. Pre-existing datasets of masked and unmasked faces are publicly available, further accelerating the dataset preparation phase. These factors collectively ensure that the project timeline is realistic and achievable.

## 8. Conclusion

Based on the analysis presented across all feasibility dimensions, MaskGuard AI is determined to be technically, operationally, and economically feasible. The system leverages proven, widely adopted technologies and frameworks to deliver an effective automated face mask detection solution. Its deployment potential spans healthcare, transport, education and commercial sectors, where continuous and consistent compliance monitoring is of paramount importance.

The project is achievable within the prescribed academic schedule and requires no proprietary or cost-prohibitive resources. MaskGuard AI therefore represents a sound, viable and impactful software engineering project with clear real world utility.

## 9. Team Collaboration

### Team Roles & Responsibilities

| Team Member | Role | Responsibilities |
|---|---|---|
| Data Fardeen | Code Development | Implemented the system, handled dataset processing, model training, and algorithm development |
| Iman Humayun | Feasibility Study | Conducted feasibility analysis and prepared the feasibility study documentation |
| Amina Asghar | Presentation Design | Prepared and organised the PowerPoint presentation |

## Contribution Matrix – Task Distribution

| Project Task | Data **Fardeen** | **Iman Humayun** | **Amina Asghar** |
|---|---|---|---|
| Dataset Preparation | ✓ | | |
| Coding & Model Implementation | ✓ | | |
| Model Training & Testing | ✓ | | |
| Feasibility Study Writing | | ✓ | |
| Documentation | | ✓ | |
| PowerPoint Preparation | | | ✓ |
| Final Review | ✓ | ✓ | ✓ |